Private Learning and Regularized Optimal Transport

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Abstract

Private data are valuable either by remaining private (for instance if they are sensitive) or, on the other hand, by being used publicly to increase some utility. These two objectives are antagonistic and leaking data might be more rewarding than concealing them. Unlike classical concepts of privacy that focus on the first point, we consider instead agents that optimize a natural trade-off between both objectives. We formalize this as an optimization problem where the objective mapping is regularized by the amount of information leaked by the agent into the system (measured as a divergence between the prior and posterior on the private data). Quite surprisingly, when combined with the entropic regularization, the Sinkhorn divergence naturally emerges in the optimization objective, making it efficiently solvable. We apply these techniques to preserve some privacy in online repeated auctions.

1 Introduction

Preserving privacy of data is a growing concern, in particular in machine learning. The classical frameworks in private learning aim at protecting sensitive data, e.g., in the medical field, where some secret information should not be publicly revealed or where the anonymity of databases must not be breached by outsiders. Algorithms must then guarantee privacy at all costs. This often leads to a large utility difference in comparison with an agent using this data without any privacy concern.

In some other situations, private data is an economic resource with a positive value that can be used publicly to optimize some utility. An agent is willing to use and reveal some private information, only if she gets a subsequent utility increase in return. The privacy is not a constraint anymore but becomes part of the objective which is then to decide how and when to use this information. This is kind of similar to a poker player deciding to bluff or not. In some situations, it might be interesting to focus solely on the utility, at the cost of a high privacy loss, while in other situations, the immediate profit for using the private data is too small, and playing independently of it (or bluffing) is better.

After a rigorous mathematical formulation of this utility/privacy trade-off, it appears that this problem can be recast as a regularized optimal transport minimization. In the specific case of entropic regularization, this problem has received a lot of interest in the recent years as it provides a computationally tractable way to approximate an optimal transport distance between distributions and is thus used in a lot of applications [13]. Our work showcases how the new Privacy-Regularized Learning problem benefits in practice from this theory.

Private Learning. Differential privacy is currently the most widely used private learning framework [15, 16, 41]. It prevents any private value of an individual to be inferred from a public aggregate information. It can be achieved by adding random noise to either the input, the output or the gradient during the learning process [10, 14, 31]. Its popularity is due to its simple, clear mathematic formulation yet with strong privacy guarantees [8]. However, its constraints are too restrictive for many problems and can lead to a low average utility [31]. Some relaxations of the differential privacy have been proposed to overcome this point [48, 7, 37, 31]. In particular, it is possible to consider

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average privacy leakage instead of ($\delta-$)worst case [17, 7], or to consider only inference from some (known) prior instead of all adjacent databases [48, 30].

The objective of private learning is to limit adversarial inference attacks. Different types of attack exist. Here, we aim at preventing attribute inference attacks [21, 53, 37], whose goal is to infer the value of private features given public observation. We thus compare the posterior and prior distributions of the private information, as done in the rational inattention literature [46, 36, 35, 18].

Optimal privacy preserving mechanisms have been recently studied for similar privacy measures, under specific conditions and strong privacy constraints [18]. Privacy can however be directly considered as a cost in the overall objective and an optimal strategy thus reveals information only if it actually leads to a significant increase in utility, whereas constrained algorithms automatically reveal as much as allowed by the constraints, without incorporating the additional cost of this revelation.

Optimal Transport. Finding an appropriate way to compare probability distributions is a major challenge in learning theory. Optimal Transport manages to provide powerful tools to compare distributions in metric spaces [51]. As a consequence, it has known an increasing interest these past years [44], especially for generative models [3, 25, 43]. However, such powerful distances often come at the expense of heavy computational costs, which are not suitable to learning algorithms. It was recently showcased that adding an entropic regularization term allows fast computation of approximated distances using Sinkhorn algorithm [47, 13]. Since then, the Sinkhorn divergence has shown promising results for applications such as generative models [24, 25], domain adaptation [12] and supervised learning [22], besides having nice theoretical properties [40, 20, 23].

Contributions and Organization of the paper. The new framework of Privacy-Regularized Learning is motivated by several applications, presented in Section 2. This problem is mathematically formulated as some optimization problem, which is convex if the privacy cost of the model belongs to some class of divergences (yet eventually in an infinite dimensional space), see Section 4. Also, if the private information space is discrete, this problem admits an optimal discrete distribution. The minimization problem then becomes dimensionally finite, but non-convex.

The mutual information between the public action and the private type is often used as a privacy measure (see [48]). If we consider this quantity as the privacy cost function, the Privacy-Regularized Learning becomes a Sinkhorn divergence minimization problem, so we are able in Section 5 to develop and use recent machinery for this problem. Finally, with a linear utility cost, the problem is equivalent to minimizing the difference of two convex functions. Using these theories, we can compare different optimization methods and illustrate the practical aspect of the model. This is done in Section 6 where we also run experiments on private learning of repeated auctions.

2 Some Applications

Our model is motivated by different applications that we describe in this section, such as online repeated auctions and learning models on servers.

2.1 Online repeated auctions

When a website wants to sell an advertisement slot, firms such as Google or Criteo take part in an auction to buy this slot for one of their client, a process illustrated in Figure 1. As this interaction happens each time a user lands on the website, this is no longer a one-time auction problem, but repeated auctions where the seller and/or the competitor might observe not just one bid, but a distribution of bids. As a consequence, if a firm were bidding truthfully, seller and other bidders would have access to its true value distribution μ. This has two possible downsides.

First, if the value distribution μ was known to the auctioneer, she could maximize her revenue at the expense of the bidder utility [1, 2, 19, 27], for instance with personalized reserve prices. Second, the auctioneer can sometimes take part in the auction and becomes a direct concurrent of the bidder (this might be a unique characteristic of online repeated auctions for ads). For instance, Google is both running some auction platforms and bidding on some ad slots for their client. As a consequence, if the distribution μ was perfectly revealed by a firm, its adversaries could use it in the future against it, by bidding more or less aggressively or by trying to conquer new markets.

It is also closely related to online pricing or repeated posted price auctions. When a user wants to buy a flight ticket (or any other good), the selling company can learn the value distribution of the buyer and can then dynamically adapt its prices in order to increase its revenue. The user can prevent this

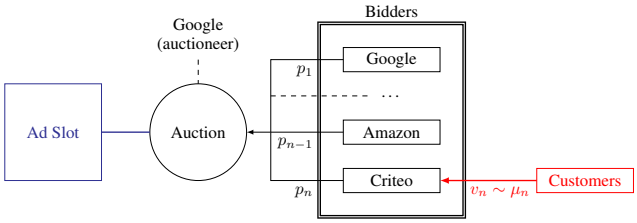


Figure 1: Online advertisement auction system.

learning in order to maximize her long term utility, even if it means refusing some apparently good offers in the short term (in poker lingo, she would be "bluffing").

2.2 Learning a model through external servers

Nowadays, several servers or clusters allow their clients to perform heavy computations remotely, for instance to learn some model parameters of a given training set (say, by training a deep neural net). The privacy concern when querying a server can sometimes be handled using homomorphic encryption [26, 5, 45], if the cluster is designed in that way (typically a public model has been learned on the server). In this case, the client sends an encrypted testing set to the server, receives new predictions from the server and recovers them for the accurate testing set. This technique, when available, is powerful, but requires heavy computations locally.

Consider instead a client wanting to learn a new model (say, a linear/logistic regression or any neural net) on a dataset that has some private component. Directly sending the training set would reveal the whole data to the server owner, besides the risk of someone else observing it. So, the agent might prefer to send slightly perturbed datasets, so that the returned parameters remain close to the accurate ones, with the data not completely revealed. If the data contain sensitive information on individuals, then differential privacy is an appropriate solution. However, it is often the case that this private part is just a single information of the client itself (say, its margin, or its current wealth, or its total number of users, etc.) that is crucial to the final model learned but should be kept more or less secret. Then differential privacy is no longer the solution, as there is only a single data to protect and/or to use. Indeed, some privacy leakage is allowed and can lead to much more accurate parameters returned by the server and a higher utility at the end; the Privacy-Regularized Learning aims at computing the best dataset to send to the server, in order to maximize the utility-privacy trade-off.

2.3 Toy Example

We finally introduce a simple toy example illustrating this privacy concept, whose formal and general model is introduced in Section 3. Suppose an agent is publicly playing an action $x \in \mathcal{X}$ to minimize a loss $x^\top c_k$, where c_k is some loss vector. The true type $k \in [K]$ is private, only known to the agent, but drawn from p_0. Without privacy concern, the agent would then solve for every k, $\min_{x \in \mathcal{X}} x^\top c_k$. Let us denote by x_k^* the optimal solution of that problem. Recall that besides maximizing her reward, the agent wants to protect her private information k (call it her "type") and that after observing the action x taken by the agent, an adversary can update her posterior distribution p_x on the hidden type.

If the agent were to play deterministically x_k^* when her type is k, then the adversary could infer the true value of k based on the played action. The agent can and should choose her action randomly to hide her true type to the adversary. So given a type k, the strategy of the agent is now a probability distribution μ_k over $\mathcal{X}$ and her expected reward is then $\mathbb{E}_{x \sim \mu_k}\left[x^\top c_k\right]$. In this case, the posterior distribution after playing the action x is computed using Bayes rule and if the different μ_k have the same support, then the posterior distribution is no longer a Dirac mass: some privacy is conserved.

The agent aims at minimizing both the objective linear loss and simultaneously the amount of information given to this adversary. A common way to measure it is given by the Kullback-Leibler (KL) divergence between the prior and the posterior [46]: $\mathrm{KL}(p_x, p_0) = \sum_{k=1}^K \log\left(\frac{p_x(k)}{p_0(k)}\right) p_x(k)$, where $p_x(k) = \frac{p_0(k)\mu_k(x)}{\sum_{l=1}^K p_0(l)\mu_l(x)}$. If the agent has some privacy concern scaling with $\lambda > 0$, the regularized loss of the agent is then $x^\top c_k + \lambda \mathrm{KL}(p_x, p_0)$ instead of $x^\top c_k$. Overall, the global objective of the agent is the minimization of $\sum_{k=1}^K p_0(k) \mathbb{E}_{x \sim \mu_k}\left[x^\top c_k + \lambda \mathrm{KL}(p_x, p_0)\right]$ over $\mu_1, \ldots, \mu_K$.

In the limit case $\lambda = 0$, the agent maximizes her reward without any privacy concern. She deterministically plays x_k^* given k. When $\lambda = \infty$, the agent only focuses on privacy and thus looks for the best

action chosen independently of the type: $x \perp\!\!\!\perp k$. It corresponds to a so called non-revealing strategy in game theory and the best strategy is then to play $\arg\min_x x^\top c[p_0]$ where $c[p_0] = \sum_{k=1}^K p_0(k) c_k$. For a positive λ, the behavior of the player will interpolate between these two extreme strategies.

This privacy preserving problem is related to repeated games with incomplete information [4], where players have private information affecting their utility functions. Playing actions leak some information to the other players, who then change their strategies in consequence. The goal is then to control the amount of information leaked to the adversaries in order to maximize its own utility. In practice, it can be impossible to compute the best adversarial strategy, because the player is often unaware of how the adversary would use this revealed information. Our framework then models this utility loss caused by adversarial actions as a function of the revealed amount of information.

3 A general model

We now introduce formally the general model sketched by the previous toy example. The agent (or player) has a private type $y \in \mathcal{Y}$ drawn according to a prior p_0 whose support can be infinite. She then chooses an action $x \in \mathcal{X}$ to maximize her utility, which depends on both her action and her type. Meanwhile, she wants to preserve the privacy of the type y. Her strategy is thus a mapping $\mathcal{Y} \to \Delta(\mathcal{X})$, where $\Delta(\mathcal{X})$ denotes the set of distributions over $\mathcal{X}$; for the sake of conciseness, we denote by $X|Y \in \Delta(\mathcal{X})^{\mathcal{Y}}$ such a strategy. In the toy example, this mapping was given by $k \mapsto \mu_k$. The adversary only observes her action x and tries to infer the type of the agent. We assume a perfect adversary, i.e., she can exactly compute the posterior distribution.

Let $c(x, y)$ be the utility loss for playing $x \in \mathcal{X}$ with the type $y \in \mathcal{Y}$. The loss due to privacy leakage corresponds to $c_{\text{priv}}(X, Y)$ where (X, Y) is the joint distribution of the action and the type. In the toy example given in Section 2.3, the utility cost was given by $c(x, k) = x^\top c_k$ and the privacy cost was the KL divergence between the prior and posterior. The privacy cost of differential privacy is

$$c_{\text{priv}}(X, Y) = \max_{(x,y,y')} \left\{ \left| \log \frac{\mathbb{P}(x \mid Y = y))}{\mathbb{P}(x \mid Y = y'))} \right| ;\ d_h(y, y') = 1 \right\}$$

where d_h is the Hamming distance [15]. The objective is to minimize the utility loss with a privacy leakage below some threshold $\varepsilon \geq 0$, i.e., minimize $\mathbb{E}_{(x,y)\sim(X,Y)}\big[c(x, y)\big]$ such that $c_{\text{priv}}(X, Y) \leq \varepsilon$. Here, the privacy leakage has some utility loss scaling with $\lambda > 0$, which can be seen as the value of information. The final objective of the agent is then to minimize the following loss:

$$\inf_{X|Y \in \Delta(\mathcal{X})^{\mathcal{Y}}} \mathbb{E}_{(x,y)\sim(X,Y)}\big[c(x, y)\big] + \lambda\, c_{\text{priv}}(X, Y). \tag{1}$$

As mentioned above, the goal is to limit the risk or the extent of type inference by the adversary. Therefore, the privacy cost is here defined as the expected value of a measure between the posterior p_x and the prior distribution p_0 of the type, i.e., $c_{\text{priv}}(X, Y) = \mathbb{E}_{x\sim X} D(p_x, p_0)$ for some function D. In the toy example of Section 2.3, $D(p_x, p_0) = \mathrm{KL}(p_x, p_0)$.

In order to have a simpler formulation of the problem, we remark that instead of defining a strategy by the conditional distribution $X|Y$, it is equivalent to see it as a joint distribution (X, Y), with a marginal over the type equal to the prior. Moreover, for a distribution $\gamma \in \Delta(\mathcal{X} \times \mathcal{Y})$, we denote by $\pi_1\#\gamma$ (resp. $\pi_2\#\gamma$) the marginal distribution of X (resp. Y): $\pi_1\#\gamma(A) = \gamma(A \times \mathcal{Y})$ and $\pi_2\#\gamma(B) = \gamma(\mathcal{X} \times B)$. The remaining of the paper focuses on the problem below, which we call **Privacy-Regularized Learning.** With the privacy cost defined as above, the minimization problem (1) is equivalent to

$$\inf_{\substack{\gamma \in \Delta(\mathcal{X}\times\mathcal{Y}) \\ \pi_2\#\gamma = p_0}} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y)\, \mathrm{d}\gamma(x, y) + \lambda \int_{\mathcal{X}\times\mathcal{Y}} D\left(p_x, p_0\right) \mathrm{d}\gamma(x, y). \tag{PRL}$$

4 An infinite dimensional convex minimization problem

In this section, we study some properties of the Problem (PRL) for specific privacy costs. Recall that D is an f-divergence, if for all distributions P, Q such that P is absolutely continuous with respect to Q, $D(P, Q) = \int_{\mathcal{Y}} f\left(\frac{\mathrm{d}P(y)}{\mathrm{d}Q(y)}\right) \mathrm{d}Q(y)$ where f is a convex function defined on $\mathbb{R}_+^*$ such that $f(1) = 0$. The set of f-divergences is large and includes common divergences such as the Kullback-Leibler divergence, the reverse Kullback-Leibler or the Total Variation distance.

Also, the min-entropy defined by $D(P, Q) = \log\left(\operatorname{ess\,sup} \mathrm{d}P/\mathrm{d}Q\right)$ is widely used in private learning [50, 48]. It corresponds to the limit of the Renyi divergence $\ln\left(\sum_{i=1}^n p_i^\alpha q_i^{1-\alpha}\right)/(\alpha - 1)$, when

$\alpha \to +\infty$ [42, 37]. Although it is not an f-divergence, the Renyi divergence derives from the f-divergence associated to the convex function $t \mapsto (t^\alpha - 1)/(\alpha - 1)$. f-divergences do not represent all the common privacy costs, but they are strongly related to many of them as we just illustrated. If we restrict the privacy costs to f-divergences, (PRL) becomes a convex minimization problem.

Theorem 1. *If D is an f-divergence,* (PRL) *is a convex problem in $\gamma \in \Delta(\mathcal{X}) \times \Delta(\mathcal{Y})$.*

The proof is given in Appendix A. Although $\Delta(\mathcal{X}) \times \Delta(\mathcal{Y})$ has generally an infinite dimension, it is dimensionally finite if both sets $\mathcal{X}$ and $\mathcal{Y}$ are discrete. A minimum can then be found using classical optimization methods such as gradient descent. In the following, D is an f-divergence.

4.1 Discrete type space

We assume here that $\mathcal{X}$ is an infinite action space and $\mathcal{Y}$ of cardinality K (or equivalently, that p_0 is a discrete prior of size K), so that it can be written as $p_0 = \sum_{k=1}^K p_0^k \delta_{y_k}$. For a fixed joint distribution γ, let the measure μ_k be defined for any $A \subset \mathcal{X}$ by $\mu_k(A) = \gamma(A \times \{y_k\})$ and $\mu = \sum_{k=1}^K \mu_k = \pi_1 \# \gamma$. The function $p^k(x) = \frac{\mathrm{d}\mu_k(x)}{\mathrm{d}\mu(x)}$, defined over the support of μ by absolute continuity, is the posterior probability of having the type k when playing x. In this specific setting, the tuple $(\mu, (p^k)_k)$ exactly determines γ. (PRL) is then equivalent to:

$$\inf_{\substack{\mu, (p^k(\cdot))_{1 \le k \le K} \\ p^k \ge 0, \sum_{l=1}^K p^l(\cdot) = 1}} \Big\{ \sum_k \int_{\mathcal{X}} \big[p^k(x) c(x, y_k) + \lambda p_0^k f\left(\frac{p^k(x)}{p_0^k}\right) \big] \mathrm{d}\mu(x);\ \forall k \le K, \int_{\mathcal{X}} p^k(x) \mathrm{d}\mu(x) = p_0^k \Big\}. \quad (2)$$

For fixed posterior distributions p^k, this is a generalized moment problem on the distribution μ [33]. The same types of arguments can then be used for the existence and the form of optimal solutions.

Theorem 2. *If the prior is dicrete of size K, for all $\varepsilon > 0$,* (PRL) *has an ε-optimal solution μ with a finite support of at most $K + 2$ points.*

The proof is delayed to Appendix A. If the support of γ is included in $\{(x_i, y_k) \mid 1 \le i \le K+2,\ 1 \le k \le K\}$, we denote it, with a slight abuse of notations, as a matrix $\gamma_{i,k} \coloneqq \gamma(\{(x_i, y_k)\})$.

Lemma 1. *In the case of a discrete prior $p_0 = \sum_{k=1}^K p_0^k \delta_{y_k}$,* (PRL) *is equivalent to:*

$$\inf_{\substack{(\gamma, x) \in \mathbb{R}_+^{(K+2) \times K} \times \mathcal{X}^{K+2} \\ \forall k \le K,\ \sum_i \gamma_{i,k} = p_0^k}} \sum_{i,k} \gamma_{i,k}\, c(x_i, y_k) + \lambda \sum_{i,k} p_0^k \times \left(\sum_{m=1}^K \gamma_{i,m} \right) \times f\left(\frac{\gamma_{i,k}}{p_0^k \sum_m \gamma_{i,m}} \right). \quad (3)$$

Theorem 2 claims that the optimization Problem (2) is equivalent to Problem (3) if we also impose no redundancy, i.e., $x_i \neq x_j$ for $i \neq j$. The proof of Lemma 1, which is delayed to the Appendix A, shows that for any redundant solution (γ, x), there is a non-redundant version with a lower value, using the subadditivity in γ. Although it seems easier to consider the dimensionally finite Problem (3), it is no longer convex in its variables. Theorem 3 gives sufficient conditions for the existence of a minimum. Its proof is delayed to Appendix A.

Theorem 3. *If the prior is discrete of size K, $\mathcal{X}$ is compact and $c(\cdot, y_k)$ is lower semi-continuous for every $k \le K$, then* (PRL) *has a minimum.*

5 Private Learning as Sinkhorn Divergence Minimization

Formally, (PRL) is expressed as a regularized Optimal Transport Minimization. In this section, we focus on the case where the privacy cost is the mutual information, as this problem will become a Sinkhorn divergence minimization with computationally tractable optimization schemes [40]. If the privacy cost is the KL divergence between the posterior and the prior, i.e., $f(t) = t \log(t)$, then the regularization term corresponds to the mutual information $I(X; Y)$. It is a widely used privacy cost [11, 9, 18] despite some limitations [48].

Recall that the Sinkhorn divergence for given distributions $(\mu, \nu) \in \Delta(\mathcal{X}) \times \Delta(\mathcal{Y})$ is defined by

$$\mathrm{OT}_{c,\lambda}(\mu, \nu) \coloneqq \min_{\gamma \in \Pi(\mu, \nu)} \int c(x, y) \mathrm{d}\gamma(x, y) + \lambda \int \log\left(\frac{\mathrm{d}\gamma(x, y)}{\mathrm{d}\mu(x) \mathrm{d}\nu(y)} \right) \mathrm{d}\gamma(x, y), \quad (4)$$

where $\Pi(\mu, \nu) = \{\gamma \in \Delta(\mathcal{X} \times \mathcal{Y}) \mid \pi_1 \# \gamma = \mu \text{ and } \pi_2 \# \gamma = \nu\}$. Problem (PRL) w.r.t. mutual information can then be rewritten as the following Optimal Transport minimization:

$$\inf_{\mu \in \Delta(\mathcal{X})} \mathrm{OT}_{c,\lambda}(\mu, p_0).$$

While the regularization term is usually only added to speed up the computations, it here directly appears in the cost of the original problem ! An approximation of $\mathrm{OT}_{c,\lambda}(\mu,\nu)$ can then be quickly computed for discrete distributions using Sinkhorn algorithm [13].

In the original problem, μ might lie in an infinite dimensional space. We instead look for an optimal distribution through a parametrized family $\mu_\theta = \sum_{j=1}^{K+2} \alpha_j(\theta)\delta_{x_j(\theta)}$ as usually done [25, 20]. The major downside is that the problem is no longer convex. In the following, p_0 and the parametrized distributions are assumed discrete. In case of continuous distributions, they could still be approximated using sampled discrete distributions as previously done for generative models [25, 23].

5.1 Minimization Algorithm

The global optimization scheme is described in Algorithm 1. The main difficulties come from the computation of the objective function and its gradient.

Sinkhorn algorithm. It was recently suggested to use the Sinkhorn algorithm, which has a linear convergence rate, to compute $\mathrm{OT}_{c,\lambda}(\mu,\nu)$ for distributions $\mu = \sum_{i=1}^{n} \alpha_i \delta_{x_i}$ and $\nu = \sum_{j=1}^{m} \beta_j \delta_{y_j}$ [32, 13]. Let K be the exponential cost matrix defined by $K_{i,j} = e^{-\frac{c(x_i,y_j)}{\lambda}}$. Any solution can be seen as a matrix in the discrete case, and the unique solution γ of the Problem (4) has the form $\mathrm{diag}(u)K\mathrm{diag}(v)$. The Sinkhorn algorithm then updates $(u,v) \leftarrow (\alpha/Kv, \beta/K^\top u)$ (with component-wise division) for n iterations or until convergence.

Algorithm 1 Optimization scheme

Input: discrete prior p_0, cost function c
Output: strategy μ_θ
Initialize θ
Until convergence do
  $\mathrm{cost} \leftarrow \mathrm{Sinkhorn}_n(\mu_\theta, p_0, c, \lambda)$
    $\{n$ iterations of Sinkhorn algorithm$\}$
  $\nabla_\theta \leftarrow \mathrm{AutoDiff}_\theta(\mathrm{loss})$
  $\theta \leftarrow \mathrm{Update}(\theta, \nabla_\theta)$ {gradient descent}
end
return μ_θ

Gradient computation. Computing $\nabla \mathrm{OT}_{c,\lambda}$ is a difficult task in Algorithm 1. A simple solution consists in using automatic differentiation, i.e., computing the gradient using the chain rule over the simple successive operations computed during the Sinkhorn algorithm [25].

The gradient can also be computed from the dual solution of Problem (4) [20, 34]. This method is faster as it does not need to store all the Sinkhorn iterations in memory and *backpropagate* through them afterwards. Convergence of Sinkhorn has yet to be guaranteed to provide an accurate approximation of the gradient (see discussion in [40]).

6 Experiments and particular cases

In this section, the case of linear utility cost is first presented and shown to have relations with DC programming. The theory of DC programming allows efficient algorithms and the performances of different possible optimization schemes are compared on a simple example. Simulations based on Algorithm 1 are then run for the real problem of online repeated auctions.

6.1 Linear utility cost

Section 4 described a general optimization scheme for Problem (PRL) with a discrete type prior. It used a dimensionally finite, non-convex problem. An objective is then to find a local minimum. Local minima can be found using classical techniques of gradient descent [52]. However in some particular cases, better schemes are possible as claimed in Section 5 for the particular case of entropic regularization. In this section, the case of a linear utility for any privacy cost is considered and we show it is related to DC programming [28]. A standard DC program is of the form $\min_{x\in\mathcal{X}} f(x) - g(x)$, where both f and g are convex functions and specific optimization schemes are then possible [49, 29, 28]. In the case of linear utility costs over a hyperrectangle, PRL can be reformulated as a DC program stated in Theorem 4. Its proof is delayed to Appendix A.

Theorem 4. *If* $\mathcal{X} = \prod_{l=1}^{d} [a_l,\ b_l]$ *and* $c(x,y) = x^\top y$*, then* (PRL) *is equivalent to the DC program*

$$\min_{\gamma\in\mathbb{R}_+^{(K+2)\times K}} \Big\{ \lambda \sum_{i,k} p_0^k h_k(\gamma_i) - \sum_{i=1}^{K+2} \| \sum_{k=1}^{K} \gamma_{i,k}\phi(y_k)\|_1 \ ;\ \forall k \le K,\ \sum_{i=1}^{K+2} \gamma_{i,k} = p_0^k \Big\},$$

with $\phi(y)^l := \frac{b_l - a_l}{2} y^l$ *and* $h_k(\gamma_i) := \left(\sum_{m=1}^{K} \gamma_{i,m}\right) f\left(\frac{\gamma_{i,k}}{p_0^k \sum_{m=1}^{K} \gamma_{i,m}}\right)$.

More generally, if the cost c is concave and the action space $\mathcal{X}$ is a polytope, optimal actions are located on the vertices of $\mathcal{X}$. In that case, we can therefore replace $\mathcal{X}$ by the set of its vertices and the problem becomes a dimensionally finite convex problem as already claimed in Section 3. Unfortunately, for some polytopes such as hyperrectangles, the number of vertices grows exponentially with the dimension and the optimization scheme is no longer tractable in large dimensions.

6.2 Comparison of convergence rates on a simple toy example

We first compare the performance of different algorithms on a simple example. We consider the linear utility loss $c(x, y) = x^\top y$ over the space $\mathcal{X} = [-1, 1]^d$ and the Kullback-Leibler divergence as privacy cost, so that both DC and Sinkhorn schemes are possible. Different methods exist for DC programming and they compute either a local or a global minimum. We choose the DCA algorithm [49] as it computes a local minimum and is thus comparable to the other considered schemes. We then compare the convergence rates of Algorithm 1, DCA and a naive gradient descent on (x, γ).

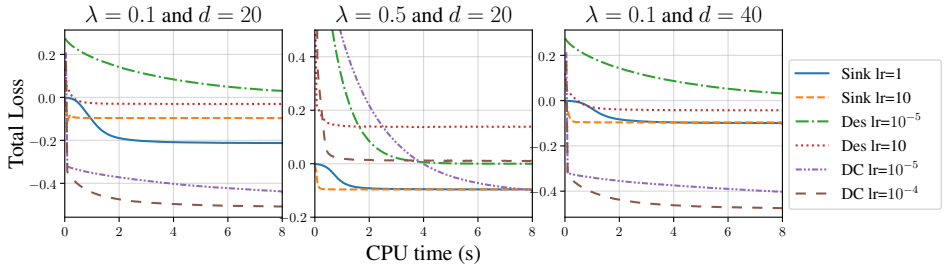


Figure 2: Comparison of optimization schemes with different learning rates (`lr`).

We optimized using projected gradient descent for different learning rates without any parameterization. The prior p_0^k is proportional to e^{Z_k}, where Z_k is chosen uniformly at random in $[0, 1]$. Each y_i is taken uniformly at random in $[-1, 1]$ and is rescaled so that $\|y_i\|_1 = 1$. The values are averaged over 200 runs. Figure 2 compares the computation times for different values of the regularization constant λ, the action space dimension d with the size of the prior $K = 100$.

The DC method finds better local minima than the other methods. This was already observed in practice [49] and confirms that it is more adapted to the structure of the problem, but only applicable in very specific cases, such as linear costs on hyperrectangles. Also, the naive descent converges to spurious local minima as it optimizes in higher dimensional spaces than the Sinkhorn method.

The Sinkhorn method seems to perform better for larger values of λ as seen in Figure 2, even if the DC technique outperforms it after some time. Indeed, given the actions, the Sinkhorn method computes the best joint distribution for each iteration and thus performs well when the privacy cost is predominant, while the DC method here computes the best actions given a joint distribution and thus performs well when the utility cost is predominant. It is thus crucial to choose the method which is most adapted to the problem structure as it can lead to significant improvement in the solution.

6.3 Utility-privacy trade-off on online repeated auctions

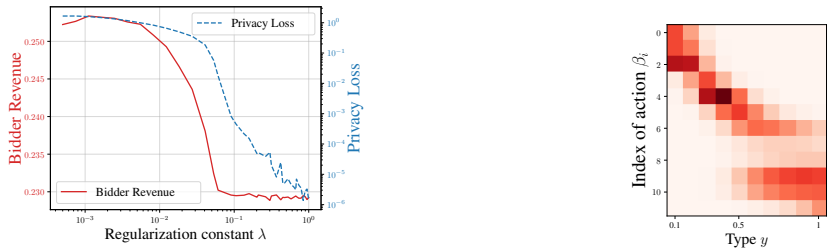
For repeated second price auctions with personalized reserve price [39], there exist numerical methods to implement an optimal strategy for the bidder [38]. For simplicity, we will only consider exponential distributions of the bidder's valuations, so that the private type y corresponds to the only parameter of this distribution, i.e., its expectation. Moreover, we assume that the prior p_0 over y is the discretized uniform distribution on $[0, 1]$ with a support of size $K = 10$; let $\{y_j\}_{j=1,\dots,K}$ be the support of p_0.

In repeated auctions, values v are repeatedly sampled from the distribution μ_{y_j} and a bidder policy is a mapping $\beta(\cdot)$ from values to bids, i.e., she bids $\beta(v)$ if her value is v. So a type y_j and a policy $\beta(\cdot)$ generate the bid distribution $\beta\#\mu_{y_j}$, which corresponds to an action in $\mathcal{X}$ in our setting. As a consequence, the set of actions of the agent is the probability distribution over $\mathbb{R}_+$ and an action ρ_i is naturally generated [44] from the valuation distribution via the optimal monotone transport map denoted by β_j^i, i.e., $\rho_i = \beta_j^i\#\mu_{y_j}$. In the particular case of exponential distributions, this implies that $\beta_i^j(v) = \beta_i(v/y_j)$ where β_i is the unique monotone transport from $\mathrm{Exp}(1)$ to ρ_i. The revenue of the bidder is then deduced for exponential distributions [38] as

$$r(\beta_i, y_j) = 1 - c(\beta_i, y_j) = \mathbb{E}_{v\sim\mathrm{Exp}(1)}\Big(\big(y_j v - \beta_i(v) + \beta_i'(v)\big)G\big(\beta_i(v)\big)\mathbb{1}_{\beta_i(v)-\beta_i'(v)\geq 0}\Big),$$

where G is the cdf of the maximum bid of the other bidders. We here consider a single non-strategic opponent with a uniform value distribution on $[0, 1]$, so that $G(x) = \min(x, 1)$.

Since the prior is discrete, Theorem 2 states that an $\varepsilon-$optimal solution has a finite support of size $K + 2$. It therefore remains to find the $K + 2$ actions that are going to be used (those actions are bid distributions, or equivalently transport maps β_i) and a joint distribution on the set of (β_i, y_j). To compute the optimal transport maps β_i, we represent and learn them as a single layer neural network of 100 ReLUs (the utility is averaged over 10^3 samples at each training time step and on 10^6 samples for the final evaluation). Figure 3 represents both utility and privacy as a function of the regularization factor λ and a typical example of joint distribution γ for a specific value of the regularization constant.



(a) Evolution of privacy-utility with λ.

(b) Joint distribution heat-map, with $\lambda = 0.01$.

Figure 3: Privacy-utility trade-off in online repeated auctions.

Naturally, both the bidder revenue and the privacy loss decrease with λ, going from revealing strategies for $\lambda \simeq 10^{-3}$ to non-revealing strategies for larger λ. They significantly drop at a critical point near 0.05. Although the bidder revenue does not significantly change (from 0.25 to 0.23), it still represents a 8% revenue difference between the non revealing strategy and the partially revealing strategy in Figure 3b. The latter randomizes the type over its possible neighbors and reveals more information when the revenue is more sensible to the action, i.e., for low types y_j here. While this strategy prevents inference on the bidder's type, the auctioneer can still infer an interval in which it is. If this kind of inference also has to be prevented, a more complex privacy cost is required.

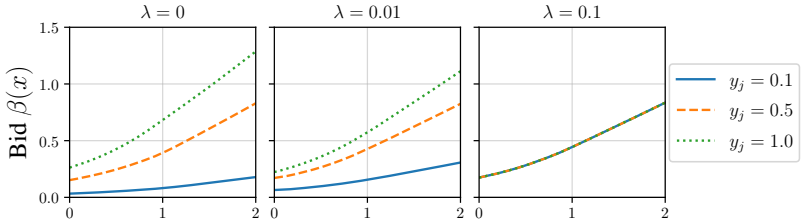


Figure 4: Evolution of the bidding strategy with the type and the regularization constant.

Figure 4 shows the most used action for different types and λ. In the revealing strategy ($\lambda = 0$), the action seems to be the same single function rescaled by the type. But as λ grows, this rescaling shrinks so that the actions perform for several types, until having a single action in the non-revealing strategy. This shrinkage is also more important for large values of y_j. This confirms the observation made above: the player loses less by hiding her type for large values than for low values and she is thus more willing to hide her type when it is large.

Besides confirming expected results, this illustrates how the Privacy-Regularized Learning framework is adapted to complex utility costs and action spaces, such as distribution or function spaces.

7 Conclusion

We formalized a new private learning concept, allowing strategies that reveal private information if it induces a significant increase in utility. For specific costs, it benefits from recent advances of Optimal Transport, which make the problem efficiently solvable for complex action spaces or utility costs. The considered privacy costs can show limitations, especially for multidimensional or metric type spaces; extending this framework to other privacy costs would have a great impact.

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A Proofs

Proof of Theorem 1. The constraint set is obviously convex. The first integral is linear and thus convex in γ. It remains to show that the privacy loss is also convex in γ. As D is an f-divergence, the privacy cost is

$$c_{\text{priv}}(\gamma) := \int_{\mathcal{X}\times\mathcal{Y}} D\left(p_x, p_0\right) \mathrm{d}\gamma(x,y) = \int_{\mathcal{X}}\int_{\mathcal{Y}} f\Big(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\gamma_1(x)\mathrm{d}p_0(y)}\Big)\mathrm{d}p_0(y)\mathrm{d}\gamma_1(x),$$

where $\gamma_1 = \pi_1\#\gamma$. For $t \in (0,1)$ and two distributions γ and μ, we can define the convex combination $\nu = t\gamma + (1-t)\mu$. By linearity of the projection π_1, $\nu_1 = t\gamma_1 + (1-t)\mu_1$. The convexity of c_{priv} actually results from the convexity of the *perspective* of f defined by $g(x_1, x_2) = x_2 f(x_1/x_2)$ [6]. It indeed implies

$$f\Big(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\gamma_1(x)\mathrm{d}p_0(y)}\Big)t\mathrm{d}\gamma_1(x) + f\Big(\frac{\mathrm{d}\mu(x,y)}{\mathrm{d}\mu_1(x)\mathrm{d}p_0(y)}\Big)(1-t)\mathrm{d}\mu_1(x) \leq f\Big(\frac{\mathrm{d}\nu(x,y)}{\mathrm{d}\nu_1(x)\mathrm{d}p_0(y)}\Big)\mathrm{d}\nu_1(x).$$

The result then directly follows when summing over $\mathcal{X}\times\mathcal{Y}$. □

Proof of Theorem 2. Let $(p^k)_k$ and μ be an ε-optimal solution of this problem. We define $g_0(x) = \sum_k \left[p^k(x)c(x,y_k) + \lambda p_0^k(x) f\left(\frac{p^k(x)}{p_0^k}\right)\right]$ and $g_k = p^k$ for $1 \leq k \leq K$. Let $\alpha_j(\mu) = \int_{\mathcal{X}} g_j \mathrm{d}\mu$ for $0 \leq j \leq K$. The considered solution μ is included in a convex hull as follows:

$$(\alpha_j(\mu))_{0\leq j\leq K} \in \mathrm{Conv}\{(g_j(x))_{0\leq j\leq K} \ / \ x \in \mathcal{X}\}.$$

So by Caratheodory theorem, there exist $K+2$ points $x_i \in \mathcal{X}$ such that $(\alpha_j(\mu))_{0\leq j\leq K} = \sum_{i=1}^{K+2} t_i (g_j(x_i))_{0\leq j\leq K}$ where (t_i) is a convex combination. Let $\mu' = \sum_{i=1}^{K+2} t_i \delta_{x_i}$. We then have $\alpha_j(\mu') = \alpha_j(\mu)$ for all j, which means that $(\mu', (p^k)_k)$ is also an ε-optimal solution of Problem (2) and the support of μ' is a subset of $\{x_i \mid i \in [K+2]\}$. □

Proof of Lemma 1. Theorem 2 claims that Problem (2) is equivalent to Problem (3) if we also impose $x_i \neq x_j$ for $i \neq j$. The value of Problem (3) is lower than the value of Problem (2) as we consider a larger feasible set. Let us consider a redundant solution (γ, x) with $x_i = x_j$ for $i \neq j$. We now show that a non redundant version of this solution will have a lower value.

Let $h_k(\gamma_i) := \left(\sum_{m=1}^K \gamma_{i,m}\right) f\left(\frac{\gamma_{i,k}}{p_0^k \sum_{m=1}^K \gamma_{i,m}}\right)$, with the conventions $f(0) = \lim_{x\to 0} f(x) \in \mathbb{R}\cup\{+\infty\}$ and $h_k(\gamma_i) = 0$ if $\sum_{m=1}^K \gamma_{i,m} = 0$.

The functions h_k are convex functions because of the convexity of the perspectives of convex functions [6]. Also, they are obviously homogeneous of degree 1. These two properties imply that the h_k are subadditive. Thus, let (γ', x') be defined by $\gamma'_{l,k} := \gamma_{l,k}$ and $x'_l := x_l$ for any $l \notin \{i,j\}$; $\gamma'_{i,k} := \gamma_{i,k} + \gamma_{j,k}$, $\gamma'_{j,k} := 0$ for any k, $x'_i := x_i$ and let x'_j be any element in $\mathcal{X} \setminus \{x_l \mid 1 \leq l \leq K+2\}$. The subadditivity of h_k implies $h_k(\gamma'_i) + h_k(\gamma'_j) \leq h_k(\gamma_i) + h_k(\gamma_j)$ for any k. The other terms in the objective function will be the same for (γ, x) and (γ', x'). It thus holds $\sum_{i,k} \gamma_{i,k} c(x_i, y_k) + \lambda \sum_{i,k} p_0^k h_k(\gamma_i) \geq \sum_{i,k} \gamma'_{i,k} c(x'_i, y_k) + \lambda \sum_{i,k} p_0^k h_k(\gamma'_i)$.

(γ', x') is in the feasible set of Problem (3) and we removed a redundant condition from x. We can thus iteratively construct a solution $(\tilde{\gamma}, \tilde{x})$ until reaching non redundancy. We then have $(\tilde{\gamma}, \tilde{x})$ in the feasible set of Problem (2) and its value is lower than the value of (γ, x). This means that allowing redundancy does not change the infimum of the problem, i.e., this yields Lemma 1. □

Proof of Theorem 3. Theorem 3 comes from the lower semi-continuity of the objective function, as claimed by Lemma 2, whose proof is given below. We recall that $h_k(\gamma_i) = \left(\sum_{m=1}^K \gamma_{i,m}\right) f\left(\frac{\gamma_{i,k}}{p_0^k \sum_m \gamma_{i,m}}\right)$.

Lemma 2. *For any k in $\{1, \dots, K\}$, h_k is lower semi-continuous.*

Let $(\gamma^{(n)}, x^{(n)})_n$ be a feasible sequence whose value converges to this infimum. As the constraint set is compact, we can assume after extraction that $(x^{(n)}, \gamma^{(n)}) \to (x, \gamma)$. As $c(\cdot, y_k)$ and h_k are all lower semi-continuous, the value of the limit is smaller than the limit value. This implies that the infimum is reached in (γ, x). □

Proof of Lemma 2. f is convex on $\mathbb{R}_+^*$. As a consequence, it is also continuous on $\mathbb{R}_+^*$. If $\lim\limits_{x \to 0^+} f(x) \in \mathbb{R}$, then f can be extended as a continuous function on $\mathbb{R}_+$ and all the h_k are thus continuous.

Otherwise, f is still continuous on $\mathbb{R}_+^*$ and $\lim\limits_{x \to 0^+} f(x) = +\infty$. Thus, h_k is continuous at γ_i as soon as $\gamma_{i,j} > 0$ for every j. If $\gamma_{i,k} = 0$, but the sum $\sum_{l=1}^K \gamma_{i,l}$ is strictly positive, then $h_k(\gamma_i) = +\infty$, but as soon as $\rho \to \gamma$, we also have an infinite limit, because $\frac{\rho_{i,k}}{p_0^k \sum_l \rho_{i,l}} \to 0$.

If $\sum_{l=1}^K \gamma_{i,l} = 0$, then $\liminf\limits_{\rho \to \gamma} f\left(\frac{\rho_{i,k}}{p_0^k \sum\limits_l \rho_{i,l}}\right) \in \mathbb{R} \cup \{+\infty\}$. As we multiply this term by a factor going to 0, $\liminf\limits_{\rho \to \gamma} h_k(\rho_i) \geq 0 = h_k(\gamma_i)$. Thus, in any case, h_k is lower semi-continuous. □

Proof of Theorem 4. Let ψ be the rescaling of $\mathcal{X}$ to $[-1, 1]^d$, i.e., $\psi(x)^l := \frac{2x^l - b_l - a_l}{b_l - a_l}$. Then, $c(x, y) = \psi(x)^T \phi(y) + \eta(y)$ where $\phi(y)^l := (b_l - a_l)\frac{y^l}{2}$ and $\eta(y) = \sum\limits_{l=1}^{d} \frac{a_l + b_l}{b_l - a_l} y^l$. Problem (3) is then equivalent to:

$$\inf_{\substack{(\gamma, x) \in \mathbb{R}^{(K+2) \times K} \times [-1,1]^{d \times (K+2)} \\ \gamma \geq 0 \\ \sum_i \gamma_{i,k} = p_0^k}} \sum_{i,k} \gamma_{i,k} (x_i^\top \phi(y_k) + \eta(y_k)) + \lambda \sum_{i,k} p_0^k h_k(\gamma_i).$$

Because of the marginal constraints, $\sum\limits_{i,k} \gamma_{i,k} \eta(y_k) = \sum\limits_k p_0^k \eta(y_k)$. This sum does not depend neither on x nor γ, so that the terms $\eta(y_k)$ can be omitted. The problem then becomes

$$\inf_{\substack{(\gamma, x) \in \mathbb{R}^{(K+2) \times K} \times [-1,1]^{d \times (K+2)} \\ \gamma \geq 0 \\ \sum_i \gamma_{i,k} = p_0^k}} \sum_i x_i^\top \left(\sum_k \gamma_{i,k} \phi(y_k) \right) + \lambda \sum_{i,k} p_0^k h_k(\gamma_i).$$

It is clear for this problem that for a fixed γ, the best x_i corresponds to $x_i^l = -\text{sign}(\sum\limits_k \gamma_{i,k} \phi(y_k)^l)$ and the term $x_i^\top \left(\sum\limits_k \gamma_{i,k} \phi(y_k) \right)$ then corresponds to the opposite of the 1-norm of $\sum\limits_k \gamma_{i,k} \phi(y_k)$, i.e., the problem is then:

$$\inf_{\substack{\gamma \in \mathbb{R}^{(K+2) \times K} \\ \gamma \geq 0 \\ \sum_i \gamma_{i,k} = p_0^k}} - \sum_i \| \sum_k \gamma_{i,k} \phi(y_k) \|_1 + \lambda \sum_{i,k} p_0^k h_k(\gamma_i).$$

□